7.6. ABSORPTION OF NEUTRONS

A beam of collimated neutrons is attenuated in a thin absorber through scattering and absorption processes in a similar manner to the attenuation of γ-rays; these processes are described in previous sections. In a thick absorber the neutrons are slowed from incident energy at the absorber face to thermal energies if the absorber is thick enough. The ultimate fate of the neutron is capture in an absorber nucleus. Because of the spread in neutron energy and the energy dependency of the capture cross-sections, no simple relation can be given for the attenuation of the neutron beam (cf. next section).

7.7. RADIATION SHIELDING

The absorption properties of nuclear radiation in different materials must be known in order to design shielding to avoid unwanted radiation effects on the surroundings by nuclear radiation sources.

For charged particles the shielding is usually slightly thicker than that required for the maximum range of the particles in the material. Absorption thicknesses of 0.2 mm are adequate to completely absorb the particles from α-decay. By contrast 15 mm of materials of low Z such as water, plastic, etc., are required for absorption of β-radiation with energies up to 3 MeV. Radiation shielding constructed from materials of higher atomic number require correspondingly thinner thicknesses. The data in Table 7.2 and the curves in Figs. 7.6 and 7.13 provide information on the thickness of absorber material required for the energy of various types of radiation.

Since γ-rays and neutrons have no definite range but exhibit a logarithmic relation between thickness and intensity, only a partial reduction of the radiation can be obtained. By combining Eqns (7.7) and (7.8), Eqn (7.27) is obtained

$$\varphi = \left(n\,A/\left(4\pi\,r^2\right)\right)\mathrm{e}^{-\mu x} \tag{7.27}$$

It is seen that the intensity from a point source of radiation can be decreased by increasing either the distance from the source r or the thickness x of the absorber. Alternately, an absorber with a higher absorption coefficient, μ, can be chosen to reduce the thickness required.

Equation (7.27) is valid only for point sources with ideal geometry, i.e. no back or multiple scattering, etc. For γ-radiation the thicker the absorber the higher is the percentage of radiation which may be scattered backwards through secondary (mainly Compton) scattering. The effect of geometry and absorber thickness can be taken into account by including a constant B in the absorption equation:

$$\varphi = B\,\varphi_0\,\mathrm{e}^{-\mu x} \tag{7.28}$$

The "*dose build-up*" *factor* B not only takes into account multiple Compton and Rayleigh scattering but also includes correction for positron formation at high γ-energies

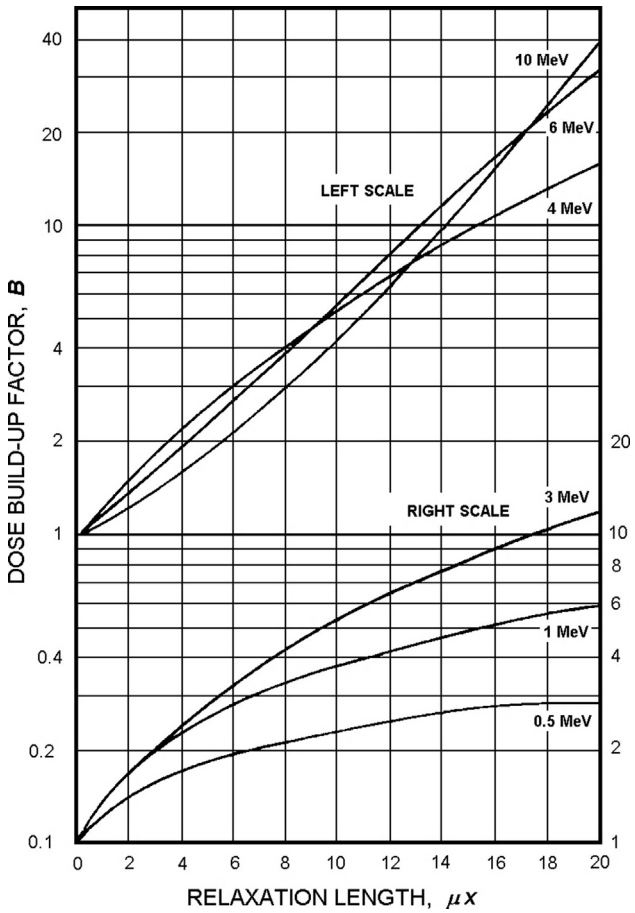


Figure 7.20 Dose build-up factors in lead for a point isotropic source. Upper curves have scale to the left, lower curves to the right. *(From Radiological Health Handbook.)*

and subsequent annihilation. Since for thick radiation shielding and high γ-energies the factor B may reach several powers of 10, it is quite important to be considered in designing biological shielding for radiation. Calculation of B is difficult and empirical data are most commonly used. Figure 7.20 gives B-values for a lead shield. The thickness of the shielding is given in relaxation lengths μx. This value is obtained from diagrams like Figure 7.17; e.g. for a 3 MeV γ, μ_m is found to be $0.046\ cm^2\ g^{-1}$. If the absorber is 0.1 m thick the linear density is $10\rho_{Pb} = 113\ g\ cm^{-2}$ and the relaxation length becomes 5.2. With Figure 7.20 this gives (for the 3 MeV γ-line) a dose build-up factor B of 3. Thus the lead shield transmits three times more radiation than is expected by the simple Eqn (7.26). The flux reduction values for concrete and lead shielding in Figure 7.15 have been adjusted to take the dose build-up into consideration.

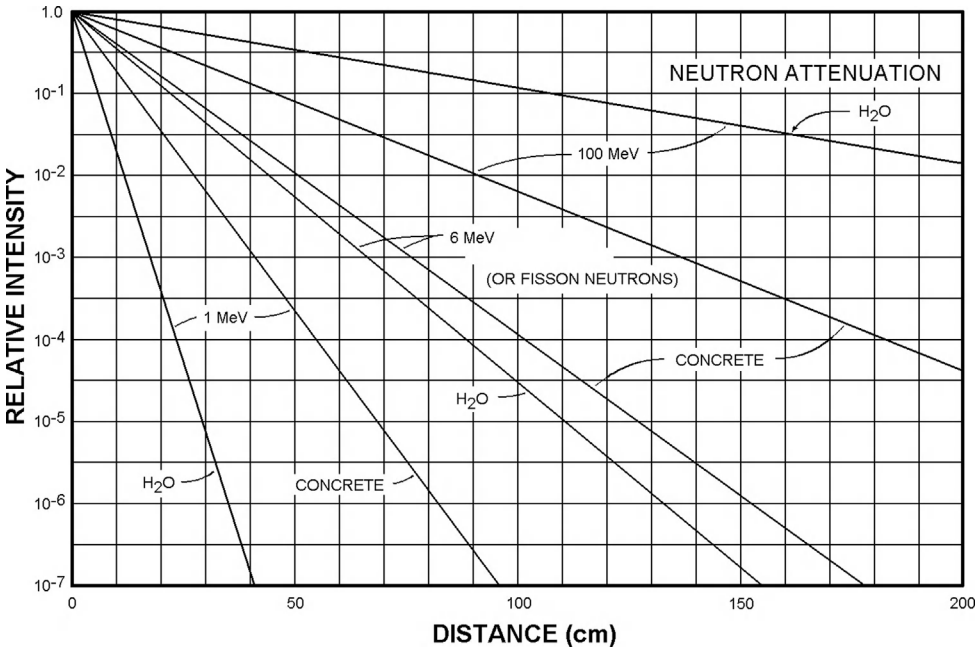


Figure 7.21 Shielding thickness necessary to reduce a neutron beam in water and concrete. The 6 MeV lines can be used for fission neutrons. *(From Nuclear Data Tables.)*

Equation (7.27) is not directly applicable for neutrons. For an estimate of required shielding we can use diagrams like that in Figure 7.21, which shows the attenuation of neutrons of three different energies in concrete and water, the most common neutron-shielding materials. It is necessary also to take into account the γ-rays emitted in neutron capture, which increases the shielding thickness required. (Please note that the order of absorption change as the energy of the neutron is increased. This is due to the energy dependence of the dominating cross sections, especially for the heavier atoms present in concrete.)

7.8. ANALYTICAL APPLICATIONS OF RADIATION ABSORPTION

In previous sections of this chapter we have shown how different particles emitted in nuclear reactions are stopped in matter without causing nuclear reactions (nuclear reactions are treated in Ch. 10 and 11). If the property of the particle is well defined (e.g. mass, charge, energy, etc) its interaction with the absorbing material can be well predicted, provided the composition of the absorbing material also is well defined. Conversely, the composition of the absorbing material (a "sample") can be determined from studying the absorption process (the "irradiation"). For example, atoms of the absorber (sample) may be knocked out and can be collected for analyses. When electrons